Charles Chow

San Jose, CA | (248) 946-7179 | yaochong.chow.20@gmail.com | yaochongchow.github.io | linkedin.com/in/charles-chow-yc

SUMMARY

Mechatronics engineer with hands-on experience in robotic hardware design, electromechanical motion systems, embedded sensor firmware, CAD, fixtures, BOMs, and hardware/software debugging. Mechanical foundation with CS/ECE graduate training across mechanical design, electrical work, control integration, Python scripting, and prototype testing.

SKILLS

Mechanical / CAD	SolidWorks, NX, Fusion 360, Ansys, FEA, GD&T, BOMs, production drawings, 3D prototyping
Mechatronics	Stepper/servo/DC motors, actuators, sensors, fixtures, test rigs, robotic parts, wiring, integration
Embedded	C/C++, Python, Zephyr, nRF52, BLE, GATT, UART, I2C, SPI, RTOS concepts, Linux
Electrical / Test	KiCad, PCB layout, Gerbers, schematics, soldering, crimping, cable harnessing, multimeter, oscilloscope basics
Systems	Control integration, hardware test plans, reliability testing, failure logs, sensor pipelines, Git, design reviews

RELEVANT EXPERIENCE

Embedded and Software Intern, NeuroLeap — May 2025 – Dec 2025

- Designed embedded C firmware on Nordic nRF52840 for 20 Hz multi-channel BLE sensor streaming, supporting validation, algorithm development, and QA workflows across 5+ engineers.
- Tuned BLE link parameters to reduce packet loss from 9.2% to 1.8%, cutting recalibration time by 60% and enabling uninterrupted 8-hour test sessions.
- Added RTT/UART instrumentation across 8 subsystems; built telemetry logging for 1.2M records/hour across 8 sensor channels.

Mechanical Engineer, Keystone Tower Systems — Jun 2022 – Mar 2023

- Designed electromechanical motion subsystems for a $2.3M automated welding system using stepper, servo, and DC motor concepts; worked with controls engineers on actuator specs and motion sequencing, achieving 15% faster cycle times.
- Built kinematic layouts enforcing +/-0.5 mm positioning tolerances across 6 configurations; resolved 3 collision scenarios through simulation-driven design review.
- Owned mechanical, controls, and field integration; led design reviews and produced documentation that cut commissioning time by 30%.

Mechanical Engineer, Engineering and Industrial Services — 2021 – 2022

- Led design direction for industrial machine fabrication projects, coordinating 6 team members across product, part, and component development.
- Created SolidWorks designs for full-scale industrial machine prototypes, including sheet metal, machined parts, fabricated assemblies, BOMs, and vendor-ready drawings.
- Produced production drawings for sheet metal, machining, and fabrication; coordinated component selection, tolerancing, and vendor feedback.

SELECT PROJECTS

Witmotion BLE IMU Streaming — C, Python, Zephyr, nRF52, BLE, SEGGER RTT

- Built Zephyr firmware on an nRF52832 DK as a BLE central for a Witmotion IMU, with GATT discovery, notification handling, reconnect scanning, and packet decoding.
- Decoded accel/gyro/orientation packets with CRC16 validation; logged timestamped CSV data with sequence, RSSI, Hz, and IMU fields.
- Added RTT host logging to reset targets, verify sample integrity, and debug packet loss, RSSI, and sample-rate stability.

ROS2 Mobile Robot Stack — ROS2, Nav2, Gazebo, Python, Linux

- Built a differential-drive ROS2 robot model in URDF/xacro and Gazebo, maintaining a stable tf2 transform tree.
- Implemented Nav2 waypoint navigation with launch-based bring-up, planner/controller tuning, rosbag2 logging, and diagnostics.
- Tuned robot bring-up, navigation, and diagnostic workflows for repeatable testing across multiple environments.

Motor Driver PCB Design — KiCad, Schematic Design, PCB Layout

- Designed a KiCad motor driver PCB for stepper and DC motor control, including schematic capture, component selection, footprints, and design rule checks.
- Produced fabrication-ready Gerbers for motor driver ICs, power regulation, decoupling, and logic-level interfacing.
- Documented motor control, power delivery, and signal-routing decisions to support review and future hardware iteration.

EDUCATION

Purdue University | M.S. Electrical and Computer Engineering (Part Time, Online) — In Progress

Northeastern University | M.S. Computer Science — May 2026

The Ohio State University | B.S. Mechanical Engineering — Dec 2020